

TASK 3 SUBMISSION

Competitor Demand Prediction Using Job Postings

NLP Preprocessing and Method Selection

Preprocessing workflow and cleaned corpus for NVIDIA

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Company specialism	NVIDIA
Task	Task 3 of 10 - NLP Preprocessing and Method Selection
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Repository	github.com/Nayab-khalid/cadetX-Competitor-Demand-Prediction

Measure	Result
Rows in / rows out	1,745 / 1,745, none dropped
Titles altered by unicode normalisation	121
Titles carrying a qualifier segment	1,201 of 1,745
Tokens per title	min 2, median 5, max 9
Distinct tokens in the vocabulary	809
Seniority coverage	662 → 1,229 (+86%)
Role function coverage	878 → 1,743 (99.9% of rows)

The constraint that shapes the method

The Task 2 dataset carries no job description text, because NVIDIA publishes through Workday and its Terms of Service prohibit automated collection. The NLP input is therefore the job title: median 6 words, maximum 13. Preprocessing built for 500-word prose would destroy that signal rather than clean it, so this pipeline parses title structure instead of reducing text.

1 Method selection

The brief allows each member to choose their own methods and libraries. Four decisions follow from the input being a short noun phrase rather than prose.

1.1 No stemming, and almost no stopwords removal

On a 500-word description, dropping function words and stemming costs nothing. On a six-word title it destroys the signal: *Solutions Architect - AI Factory Deployment* has no words to spare. A standard stemmer on this vocabulary produces *analysi*, *architectur* and *seri*, which are worse than the plurals they replace.

Instead the pipeline uses a controlled singularisation map of 26 entries covering the plurals that actually occur (tools to tool, systems to system, libraries to library), and a stopwords list of 11 genuine function words plus bare years such as 2027.

1.2 Structural parsing rather than bag-of-words

NVIDIA titles follow a consistent grammar:

```
[seniority] [role] , - - [team / product / domain qualifier]

Senior      Software Engineer , AI Developer Tools - Cloud Software
Principal   Firmware Engineer - Data Center Server Management
            Solutions Architect, Supercomputing
```

1,201 of 1,745 titles carry a separator, and the head noun of the first segment is drawn from a small set: Engineer 964, Architect 314, Manager 313. Parsing that structure yields far more than counting words does.

1.3 Fill the gaps the source left

The raw feed populated seniority on 662 rows and job category on 878. Both are largely recoverable from the title, and doing so is the single largest quality gain available here.

1.4 Libraries

pandas, plus re and unicodedata from the standard library. spaCy and NLTK are not used, and not merely because they were unavailable in the environment: both are built for sentence-level text, their taggers and lemmatisers are trained on prose, and a six-word noun phrase gives them almost nothing to work with. A rule set that is readable, auditable and reviewable by a teammate is worth more here than a dependency that adds no accuracy. Every rule lives in a named constant at the top of the script.

Out of scope by design: skill and technology extraction is Task 4. This task produces cleaned text and structural fields only, so the boundary between the two stays clean.

2 The pipeline

#	Stage	What it does
1	Read	Loads the Task 2 raw dataset
2	Unicode normalisation	NFKC, then maps typographic characters to ASCII: 91 en dashes, 19 em dashes, 3 non-breaking spaces, 1 non-breaking hyphen. Without this, <i>Engineer – Cloud</i> and <i>Engineer - Cloud</i> tokenise differently
3	Segment	Splits on comma, hyphen, en dash, em dash and pipe into a core role and its qualifiers

#	Stage	What it does
4	Role function	Matches the head noun of the core against 24 patterns, falling back to the full title
5	Seniority	Matches 9 ordered rank patterns; most specific wins, so <i>Senior Director</i> resolves to Director
6	Clean text	Strips rank words and punctuation, lowercases, collapses whitespace. Plus, hash and hyphen are preserved for C++, C# and multi-gpu
7	Tokenise	Splits, drops stopwords and bare years, applies controlled singularisation
8	Flag repeats	Counts identical titles; 1,463 of 1,745 titles are unique

Stage 4 runs **before** stage 5 on purpose: seniority needs to know whether *Lead* is this row's head noun. Section 4 explains why.

2.1 Fields produced

Added to the 25 columns carried through from Task 2:

Field	Description
cleaned_description	The cleaned title text. Team schema field name , used so cross-company joins in Task 6 work unchanged. For NVIDIA it derives from the title, not a description
cleaned_title_core	The cleaned core role, with qualifiers removed
title_qualifiers	JSON list of the team / product / domain segments
n_qualifiers	How many qualifier segments the title carried
role_function	Engineer, Architect, Manager, Lead, Scientist and so on
seniority_derived	Intern, Junior, Senior, Staff, Lead, Principal, Distinguished, Director, Executive, Unspecified
title_tokens	JSON list of cleaned tokens, the NLP input for Task 4
n_tokens	Token count
title_repeat_count	How many postings share this exact title

3 Results

Seniority	Postings	Role function	Postings
Senior	1,081	Engineer	964
Unspecified	516	Architect	314
Principal	57	Manager	313
Junior	38	Lead	24
Director	19	Scientist	23
Intern	15	Developer	20
Distinguished	10	Director	18
Lead	6	Researcher	15
Staff	3	Other	2

62 per cent of NVIDIA's open roles are explicitly Senior or above. The 516 unspecified are titles that genuinely carry no rank word, such as *Solutions Architect*, *Supercomputing*; they are left unspecified rather than guessed at.

3.1 Most frequent tokens

Token	Count	Share	Token	Count	Share
engineer	971	55.6%	developer	116	6.7%
software	486	27.9%	infrastructure	98	5.6%
ai	330	18.9%	learning	90	5.2%
manager	319	18.3%	networking	88	5.0%
architect	317	18.2%	performance	81	4.6%
solution	211	12.1%	platform	77	4.4%
system	197	11.3%	verification	77	4.4%
design	129	7.4%	data	76	4.4%
cloud	118	6.8%	program	77	4.4%

The full table of all 809 tokens is in `nvidia_token_frequency_20260902.csv`.

4 Validation, and two bugs it caught

A 12-row random sample and a full review of the Other bucket were checked by hand. That found two real errors, both now fixed.

Bug 1: Lead was read as a rank when it is a function

NVIDIA has a family of roles whose head noun is Lead or Leader: Global Account Leader Automotive, Segment Sales Leader, Business Development Lead. The rank stripper removed the word, leaving those titles with no role noun, so 24 postings fell into Other and their seniority was inflated to Lead. Fixed by matching role function first and suppressing the Lead rank rule when Lead is the head noun, and by removing lead and leader from the words stripped out of the cleaned text.

Bug 2: new-graduate postings were not detected as junior

The pattern required "new grad" adjacently, so "New College Grad 2027" and "NVIDIA 2027 New College Graduate" were missed. Junior rose from 11 to 38 once fixed.

After both fixes, Other fell from 30 rows to 2. Those 2 are genuine graduate-programme postings with no role noun at all, and are left as Other rather than forced into a category.

5 Limitations

#	Limitation
1	A title is not a description. This is the ceiling on everything downstream. Five tokens per title cannot carry what 500 words would.
2	516 rows have no seniority signal. Not an error, just absent from the source text. They are marked Unspecified, and any analysis that splits by seniority must say what it does with them.

#	Limitation
3	Rules are tuned to NVIDIA's title conventions. Another company's titles would need their own review. This is a per-company pipeline, not a general one.
4	role_function is not a job family. It is the head noun: a Solutions Architect in sales and a Silicon Architect both read as Architect. Grouping into families belongs in Task 4.
5	Qualifier segments are kept as raw text. Normalising AI Infra, AI Infrastructure and AI Factory onto shared concepts is Task 4's job, not this one's.

6 Two points for the team

1. **The seniority vocabulary needs extending.** docs/data-schema.md lists Junior, Mid, Senior, Lead and Principal. NVIDIA's titles also carry Intern, Staff, Distinguished, Director and Executive. I have used the extended set and flagged it here rather than forcing real values into five buckets. It needs ratifying so all four members use the same list.
2. **cleaned_description means different things per member.** For members with description text it is preprocessed prose; for NVIDIA it is a preprocessed title. The column name is shared so the Task 6 joins work, but anything comparing text length, vocabulary size or token counts across companies will be comparing incomparable things unless it accounts for this.

Files

File	Rows	Contents
data/processed/nvidia_postings_clean_20260902.csv	1,745	Cleaned dataset, 34 columns
data/processed/nvidia_token_frequency_20260902.csv	809	Token frequency across the corpus
preprocess_nvidia.py	-	The pipeline, re-runnable from a clean checkout

```
cd work/task-3/nvidia-nayab-khalid
python preprocess_nvidia.py
```

Declaration	No personal or sensitive data is present in the cleaned corpus. No new data was collected for this task; the input is the dataset committed in Task 2.
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